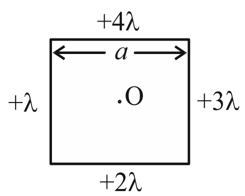
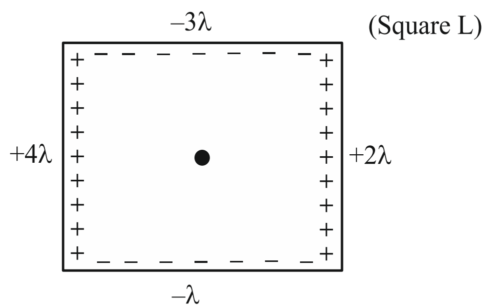
Lakshya JEE 2027

KPP - 03 | **Electrostatics** | **BY: Saleem Bhaiya**

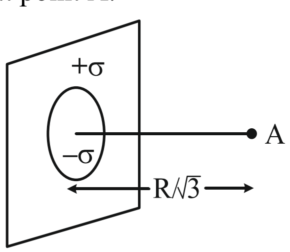
1. Four uniformly charged wires of length a are arranged to form a square. Linear charge density of each wire is as shown. Electric field intensity at centre of square is $\frac{nk\lambda}{a}$ then value of n.



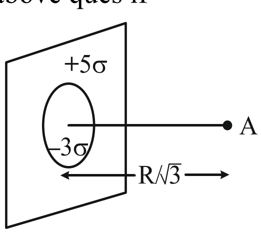
2. Find E.F. at point A.



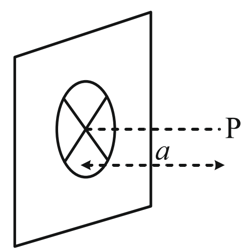
3. From a ∞ sheet (+σ), a disc of radius 'R' is removed and in emptiy part another disc of charge density –σ is placed.
Find E.F. at point A.



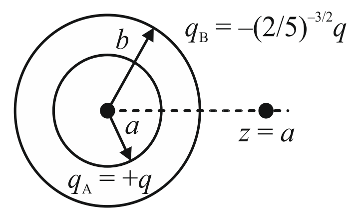
4. Repeat the above ques if



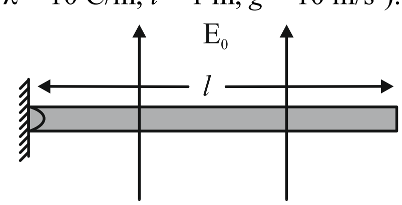
5. An infinitely long non-conducting plane of charge density σ has circular aperture of certain radius carved out from it. The electric field at a point which is at a distance 'a' from the centre of the aperture is $\sigma/2\sqrt{2}\epsilon_0$. Find the radius of aperture.



6. Two concentric rings, one of radius 'a' and the other of radius 'b' have the charges $+q$ and $-(2/5)^{-3/2}q$ respectively as shown in the figure. Find the ratio b/a if a charge particle placed on the axis at $z = a$ is in equilibrium.



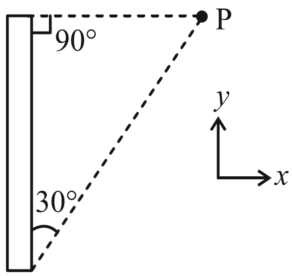
7. A thin insulating uniformly charged (linearly charged density λ) rod is hinged about one of its ends. It can rotate in vertical plane. If rod is in equilibrium by applying vertical electric field E as shown in figure. Find the value of E(in N/C). (Given that mass of rod 2 kg, $\lambda = 10$ C/m, $l = 1$ m, $g = 10$ m/s^2).



8. Five positive equal charges are placed at vertices of a regular hexagon and net electric field at the centre is E_1. A negative charge having equal magnitude is placed sixth vertex and then net electric field is E_2. Find $\frac{E_2}{E_1}$.

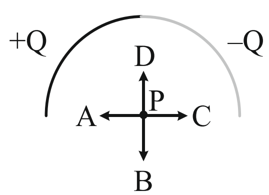
(1) 2 (2) 1
(3) 3 (4) None of these

9. The direction (θ) of $\vec{E}$ at point P due to uniformly charged finite rod will be:



(1) at angle 30° from x-axis
(2) 45° from x-axis
(3) 60° from x-axis
(4) none of these

10. As shown in the figure to the right, an insulating rod is set into the shape of a semicircle. The left half of the rod has a charge of +Q uniformly distributed along its length, and the right half of the rod has a charge of –Q uniformly distributed along its length. What vector shows the correct direction of the electric field at point P, the centre of the semicircle?



(1) A (2) B
(3) C (4) D

11. A nonconducting ring of radius R has uniformly distributed positive charge Q. A small part of the ring, of length d, is removed ($d << R$). The electric field at the centre of the ring will now be:

(1) directed towards the gap, inversely proportional to R^3.
(2) directed towards the gap, inversely proportional to R^2.
(3) directed away from the gap, inversely proportional to R^3.
(4) directed away from the gap, inversely proportional to R^2.

12. A particle of mass m, charge –Q is constrained to move along the axis of a ring of radius a. The ring carries a uniform charge density $+\lambda$ along its circumference. Initially, the particle lies in the plane of the ring at a point where no net force acts on it. The period of oscillation of the particle when it is displaced slightly from its equilibrium position is:

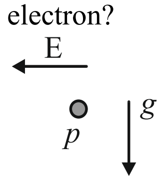
(1) $T = 4\pi\sqrt{\frac{\varepsilon_0 m a^2}{\lambda Q}}$

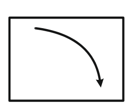
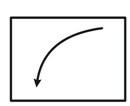
(2) $T = 2\pi\sqrt{\frac{2\varepsilon_0 m a^2}{\lambda Q}}$

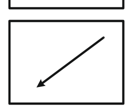
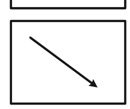
(3) $T = 2\pi\sqrt{\frac{4\varepsilon_0 m a^2}{\lambda Q}}$

(4) $T = 2\pi\sqrt{\frac{\varepsilon_0 m a^2}{2\lambda Q}}$

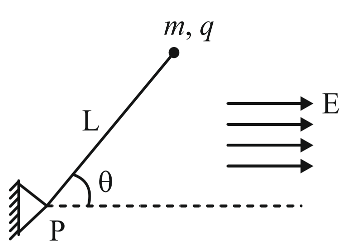
13. A negatively charged particle p is placed, initially at rest, in a constant, uniform gravitational field and a constant, uniform electric field as shown in the diagram. What qualitatively, is the shape of the trajectory of the electron?



(1)  (2) 

(3)  (4) 

14. A particle of mass m and charge q is attached to a light rod of length L. The rod can rotate freely in the plane of paper about the other end, which is hinged at P. The entire assembly lies in a uniform electric field E also acting in the plane of paper as shown. The rod is released from rest when it makes an angle θ with the electric field direction. Determine the speed of the particle when the rod is parallel to the electric field.



(1) $\left(\frac{2qEL(1-\cos\theta)}{m}\right)^{1/2}$

(2) $\left(\frac{2qEL(1-\sin\theta)}{m}\right)^{1/2}$

(3) $\left(\frac{qEL(1-\cos\theta)}{2m}\right)^{1/2}$

(4) $\left(\frac{2qEL\cos\theta}{m}\right)^{1/2}$

15. Under the influence of the Coulomb field of charge +Q, a charge $-q$ is moving around it in an elliptical orbit. Find out the correct statement(s).

[IT-JEE 2009]

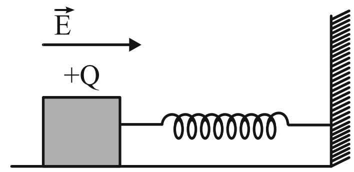
(1) The angular momentum of the charge $-q$ is constant

(2) The linear momentum of the charge $-q$ is constant

(3) The angular velocity of the charge $-q$ is constant

(4) The linear speed of the charge $-q$ is constant

16. A wooden block performs SHM on a frictionless surface with frequency, v_0. The block carries a charge +Q on its surface. If now a uniform electric field $\vec{E}$ is switched-on as shown, then the SHM of the block will be: **[IIT-JEE 2011]**



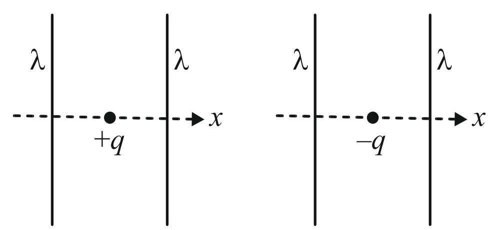
(1) of the same frequency and with shifted mean position

(2) of the same frequency and with the same mean position

(3) of changed frequency and with shifted mean position

(4) of changed frequency and with the same mean position

17. The figures below depict two situations in which two infinitely long static line charges of constant positive line charge density λ are kept parallel to each other. In their resulting electric field, point charges q and $-q$ are kept in equilibrium between them. The point charges are confined to move in the x direction only. If they are given a small displacement about their equlibrium positions, then the correct statement(s) is (are): **[JEE-Advance-2015]**



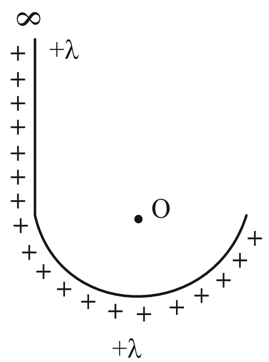
(1) Both charges execute simple harmonic motion

(2) Both charges will continue moving in the direction of their displacement

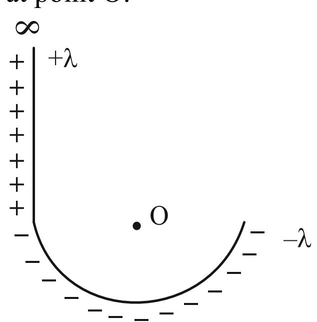
(3) Charge $+q$ executes simple harmonic motion while charge $-q$ continues moving in the direction of its displacement.

(4) Charge $-q$ executes simple harmonic motion while charge $+q$ continues moving in the direction of its displacement.

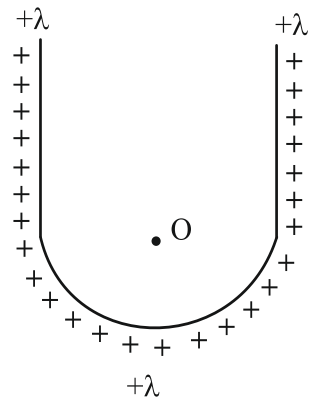
18. Find E.F. at point O.



19. Find E.F. at point O.



20. Find E.F. at point O.



Answer Key

1. (8)
2. (*)
3. (*)
4. (*)
5. (*a*)
6. (2)
7. (2)
8. (1)
9. (1)
10. (3)
11. (1)
12. (2)
13. (4)
14. (1)
15. (1)
16. (1)
17. (3)
18. (*)
19. (*)
20. (*)